

Question	Marking Guidance	Mark	Comments
10.1	<u>Starch (max 3)</u> 1. Helical/ spiral shape so compact; 2. Large (molecule)/insoluble so osmotically inactive; 3. Branched so glucose is (easily) released for respiration; 4. Large (molecule) so cannot leave cell/cross cell-surface membrane; <u>Cellulose (max 3)</u> 5. Long, straight/unbranched chains of β glucose; 6. Joined by hydrogen bonding; 7. To form (micro/macro)fibrils; 8. Provides rigidity/strength;	5 max	2. Accept: does not affect water potential/Ψ. 3. Ignore: unbranched. 6. Note: references to 'strong hydrogen bonds' disqualifies this mark point.
10.2	1. (At source) sucrose is actively (transported) into the phloem/sieve element/tube; 2. By companion/transfer cells; 3. Lowers water potential in phloem/sieve element/tube and water enters by <u>osmosis</u>; 4. (Produces) high (hydrostatic) pressure; 5. <u>Mass</u> flow/transport towards sink/roots/storage tissue; 6. At sink/roots sugars are removed/unloaded;	5 max	1. Accept: 'sugar/s' for sucrose but reject other named sugars e.g. glucose. 1. Accept: co-transport (with H^+ ions). 4. Accept: pressure gradient. 5. Accept: sieve element/tube. 6. Accept: at sink/roots sugars are used in respiration/stored.